Project Design Phase
Problem – Solution Fit Template

Date	27 February 2026
Team ID	TSK841298
Project Name	HematoVision – Advanced Blood Cell Classification Using Transfer Learning
Maximum Marks	2 Marks

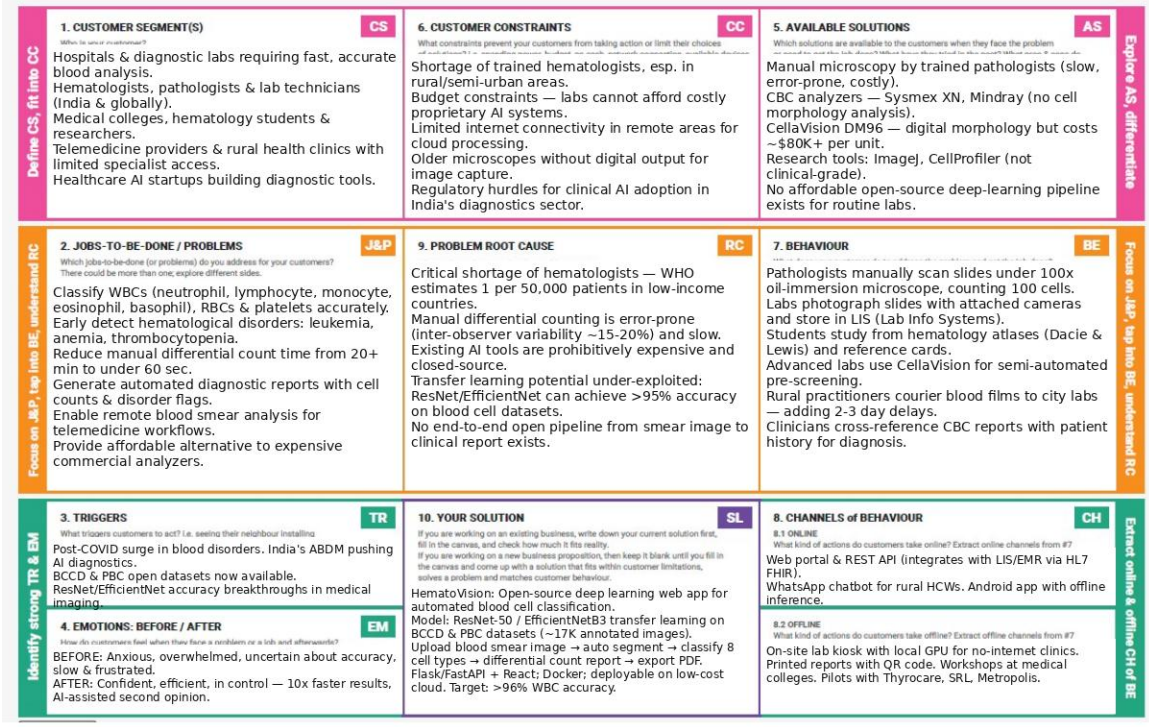
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Hospitals & diagnostic labs requiring fast, accurate blood analysis.
Hematologists, pathologists & lab technicians (India & globally).
Medical colleges, hematology students & researchers.
Telemedicine providers & rural health clinics with limited specialist access.
Healthcare AI startups building diagnostic tools.

6. CUSTOMER CONSTRAINTS CC

What constraints prevent your customers from taking action or limit their choices

Shortage of trained hematologists, esp. in rural/semi-urban areas.
Budget constraints — labs cannot afford costly proprietary AI systems.
Limited internet connectivity in remote areas for cloud processing.
Older microscopes without digital output for image capture.
Regulatory hurdles for clinical AI adoption in India's diagnostics sector.

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem

Manual microscopy by trained pathologists (slow, error-prone, costly).
CBC analyzers — Sysmex XN, Mindray (no cell morphology analysis).
CellaVision DM96 — digital morphology but costs ~$80K+ per unit.
Research tools: ImageJ, CellProfiler (not clinical-grade).
No affordable open-source deep-learning pipeline exists for routine labs.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

Classify WBCs (neutrophil, lymphocyte, monocyte, eosinophil, basophil), RBCs & platelets accurately.
Early detect hematological disorders: leukemia, anemia, thrombocytopenia.
Reduce manual differential count time from 20+ min to under 60 sec.
Generate automated diagnostic reports with cell counts & disorder flags.
Enable remote blood smear analysis for telemedicine workflows.
Provide affordable alternative to expensive commercial analyzers.

9. PROBLEM ROOT CAUSE RC

Critical shortage of hematologists — WHO estimates 1 per 50,000 patients in low-income countries.
Manual differential counting is error-prone (inter-observer variability ~15-20%) and slow.
Existing AI tools are prohibitively expensive and closed-source.
Transfer learning potential under-exploited: ResNet/EfficientNet can achieve >95% accuracy on blood cell datasets.
No end-to-end open pipeline from smear image to clinical report exists.

7. BEHAVIOUR BE

Pathologists manually scan slides under 100x oil-immersion microscope, counting 100 cells.
Labs photograph slides with attached cameras and store in LIS (Lab Info Systems).
Students study from hematology atlases (Dacie & Lewis) and reference cards.
Advanced labs use CellaVision for semi-automated pre-screening.
Rural practitioners courier blood films to city labs — adding 2-3 day delays.
Clinicians cross-reference CBC reports with patient history for diagnosis.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS TR

What triggers customers to act? i.e. seeing their neighbour installing

Post-COVID surge in blood disorders. India's ABDM pushing AI diagnostics.
BCCD & PBC open datasets now available.
ResNet/EfficientNet accuracy breakthroughs in medical imaging.

4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards?

BEFORE: Anxious, overwhelmed, uncertain about accuracy, slow & frustrated.
AFTER: Confident, efficient, in control — 10x faster results, AI-assisted second opinion.

10. YOUR SOLUTION SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

HematoVision: Open-source deep learning web app for automated blood cell classification.
Model: ResNet-50 / EfficientNetB3 transfer learning on BCCD & PBC datasets (~17K annotated images).
Upload blood smear image → auto segment → classify 8 cell types → differential count report → export PDF.
Flask/FastAPI + React; Docker; deployable on low-cost cloud. Target: >96% WBC accuracy.

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE
What kind of actions do customers take online? Extract online channels from #7

Web portal & REST API (integrates with LIS/EMR via HL7 FHIR).
WhatsApp chatbot for rural HCWs. Android app with offline inference.

8.2 OFFLINE
What kind of actions do customers take offline? Extract offline channels from #7

On-site lab kiosk with local GPU for no-internet clinics.
Printed reports with QR code. Workshops at medical colleges. Pilots with Thyrocare, SRL, Metropolis.

Extract online & offline CH of BE

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe